MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY

An Autonomous Institution

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Accredited by NBA, New Delhi and NAAC with 'A' Grade
Kalitheerthalkuppam, Puducherry- 605 107.

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INTERNAL TEST-I

SUB CODE/NAME: CSPC403 DESIGN &ANALYSIS OF ALGORITHMS

YEAR/SEM/SEC:II/IV/B **TIME: 8.50AM TO10.30 AM**

DATE: 10.02.2026 **MARKS: 50**

CO1	Analyze the performance of various sorting algorithms such as heap ,radix sort, bubble sort and insertion sort.
CO2	Explain the concepts of Divide and Conquer method and the Greedy method regarding their approach to problem-

[Bloom's Taxonomy: K1- Remember, K2- Understand ,K3-Apply, K4-Analysis]

PART-A

ANSWER ALL THE QUESTIONS **(10X2=20)**

1. Define an algorithm. (CO1 ,K1)
2. List out the properties of Asymptotic notation (CO1, K2)
3. Compute an algorithm for Merge sort. (CO1,K2)
4. Classify the time complexity of all sorting algorithm. (CO1 ,K2)
5. Define searching and its types. (CO1 ,K1)
6. Compute the performance analysis of an algorithm. (CO2 ,K2)
7. Define the characteristic of an algorithm . (CO2 ,K1)
8. Compare linear search and binary search (CO1 ,K2)
9. Differentiate between feasible solution and optimal solution (CO2,K2)

10.Outline the principle of divide and conquer technique *(CO2 ,K2)*

PART-B

ANSWER ALL QUESTIONS **(3X10=30)**

11. Write algorithm with neat sketch with example **(CO1,K3)**

a)Selection sort (12,24,8,71,4,23,6) **(5)**

b)Binary search(n=10 and 7,12,45,67,78,89,95,120,156,430 key searched element is 95 **(5)**

12 . Describe about Strassen's Matrix Multiplication and Strassen's Formula. Solve this matrix using strassen's formula. **(CO2,K3)**

$$A=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad B=\begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$$

(10)

13. Analyze Quick sort algorithm :12,78,11,4,23,55,19,29,31 **(CO2,K4)**

(10)

******** All the best ********